C2000Ware v26.01.00.00.STS Release Notes

1. Introduction

C2000Ware is a cohesive set of development tools for C2000 real-time controllers. It includes device-specific drivers, bit-fields, libraries (math, DSP, Control, Signal Generation), peripheral examples, utilities, hardware files, and documentation. Application specific software and hardware files are delivered through additional Software Development Kits (SDK).

2. Device Support

The following devices are supported with this release

- MCU's
    - F28P551x
    - F28E12x
    - F28P55x
    - F28P65x
    - F280015x
    - F280013x
    - F28003x
    - F28002x
    - F2838x
    - F28004x
    - F28x7x

  - Prior to F28x7x
- Hardware Platforms
  - ControlCARDs and LaunchPAD's

3. What Is Supported

- Driverlib:
  - Driverlib for F28P551x, F28E12x, F28P55x, F28P65x, F280015x, F280013x, F28003x, F28002x and F2838x - All content in EABI only
  - Driverlib for C28x and CM (F2838x) peripherals
  - F28004x and F28x7x - All contents in both COFF and EABI. Recommendation is to use EABI for all new development.
- SysConfig:
  - Support for Control Peripherals
  - Support for Analog Peripherals
  - Support for System Peripherals
  - Support for Communication Peripherals
  - Support for C2000 and Third-Party Libraries
  - Support for Board View (Launchpad and ControlCard)
  - Support for Custom Board Migration
  - Support for Device to Device Auto-Migration
- Universal Project Support Examples (Easy migration support across C2000 using SysCfg)
  - Universal , timer_ex1, led_ex1_blinky, gpio_ex2_toggle, i2c_ex1_loopback, spi_ex1_loopback, spi_ex2_loopback_fifo_interrupts, spi_ex3_external_loopback, spi_ex4_external_loopback_fifo_interrupts, spi_ex5_loopback_dma, sci_ex1_loopback, baud_tune_via_uart, lin_ex5, dma_ex1_gsram_transfer
- Bitfields :
  - Headers and examples for F28P551x, F28E12x, F28P55x, F28P65x, F280015x, F280013x, F28003x, F28002x and F2838x - All content in EABI only
  - F28004x and F28x7x - In COFF only
- Kernel
  - FreeRTOS kernel for C28x and demos on F28P551x, F28E12x, F2838x, F28002x, F28003x , F280013x, F280015x , F28P65x ,F28P55x, F2837xs, F28004x
- EtherCAT:
  - CPU1/CM EtherCAT HAL drivers and test examples for F2838x and F28P65x
  - CPU1/CM Stack Configuration file for Beckhoff's Slave Stack Code Tool for F2838x and F28P65x
  - CPU1/CM Slave Stack demo examples for F2838x and F28P65x

  - EtherCAT - FoE example for C28x and CM
- FPU DSP and FastRTS Library
  - Single Precision FPU DSP library consisting of various optimized FFT, filter, math, and vector assembly routines
  - Double Precision FPU DSP library consisting of various optimized FFT, filter and vector assembly routines
  - Examples demonstrating the usage of single (F28002x/F2838x) and double precision (F2838x) FPU DSP assembly routines including those running with ADC
  - Single and double precision FastRTS libraries consisting of various optimized math routines
  - Examples demonstrating the usage of single (F28002x/F2838x) and double (F2838x) precision FastRTS math routines
  - RAM, FLASH and ROMTABLE configurations are supported for examples
  - F2837xD examples are also supported same as in the previous release without any modification.
- IQMath Library:
  - Examples on F2837x – COFF and EABI support
  - Examples on F28002x, F2838x , F28003x , F280013x , F280015x , F28P65x , F28P55x and F28E12x- EABI support
- FASTINTDIV :
  - F28003x, F28002x, F2838x examples to showcase the usage of fast division intrinsic
- Scale Factor Optimization (SFO) Library
  - Migrated to use driverlibs instead of bitfields
- Fixed Point DSP Library
- VCRC Library and examples
  - Examples demonstrating library capabilities
- Digital Control Library (DCL)
  - Examples demonstrating library capabilities
- Crypto Software Library
  - Publicly available software AES, SHA256, SHA384, RSA and SHA512 .
- AI examples on F28P55x, F28P65x, F28003x, F28004x
  - Arc Fault
  - Blower Imbalance
  - Fan Blade Anomaly Detection
  - Generic Timeseries - Anomaly Detection, Classification , Regression, Forecasting
  - Motor Fault
  - Torque Measurement
  - Forecasting PMSM Motor Temp
  - HVAC Indoor Temp Forecast

  - Washing machine load weighing
- PMBus slave stack
  - F28P551x, F28P55x, F28P65x, F280015x, F28003x, F28004x, F2838x, F28002x: Master/slave example demonstrating PMBus v1.2 capability
  - PMBus over I2C support for F28P551x, F28P55x, F28P65x, F280015x, F28003x, F28004x, F2838x, F28002x.
- Flash API
- Tools and Utilities
  - SysConfig support for C2000 analog and digital modules
  - CLB Tool for F28P551x, F28P55x, F28P65x, F28003x, F28002x, F2838x, F28004x and F28x7x
  - DCSM security tool
  - Linker CMD Tool
  - MCU Control Center - Data Transfer Tool (BETA)
  - MCU Control Center - MCU Signal Sight (BETA)
  - MCU Control Center - MCU Clinic (BETA)
- Software Diagnostic Library
  - Library and examples for F28003x, F28004x, F28002x, F2838x, F280013x, F280015x, F28P65x , F28P55x, F28E12x and F28P551x.
  - Demonstrates software diagnostics and software tests of hardware diagnostics
- Bug fixes and enhancements

Note: The minimum version of Code Composer Studio required is CCSv20.4.1 and compiler version is 22.6.3.LTS. Please refer to the Dependencies section for the download links.

The validation has been carried out using these tool versions and it is recommended to use these tool versions.

4. What Is Not Supported

- In general, features or drivers not mentioned as part of "What Is Supported" section are not supported in this release.
- Driverlib & bitfields:
  - Bitfield examples in EABI for F28x7x and F28004x.
- SGEN

5. New In This Release

- Support for F28P551x device - driverlib, bit field headers and examples
- Safety Diagnostic Library on F28P551x
- FreeRTOS demos on F28P551x
- Libraries updated to add support for F28P551x
  - FPU DSP
  - IQMath Library
  - Fixed Point DSP Library
  - Digital Control Library (DCL)
  - PMBus Slave Communication Stack Library
  - VCRC Library
  - CLAmath Library
- SysConfig support for F28P551x
- DCSM Tool and example for F28P551x
- Boot ROM for F28P551x:
  - Source code and symbol libraries
- Academy examples for F28P551x
- Updated Flash API for F28P551x
- Update structure for C28x crypto libs
  - libraries/security/crypto/c28/

```
├── aes/ - AES encryption
├── rsa/ - RSA encryption
├── ecdsa/ - ECDSA signatures
├── sha/ - SHA hashing
├── middleware/ - mcuboot core files
├── examples/ - Device wise examples
├── signing_tools/ - mcuboot signing scripts
└── docs/ - Documentation
```

- Asymmetric Secure Boot for F28P55x
  - Supports Mcuboot based secure boot with ECDSA384 signature.
- MCUBoot based Asymmetric Secure Firmware Update for F28P55x(ECDSA384 signature based).
- Addition of mcuboot secure boot signing tools
- Clinic/transfer tool enhancements
- New Examples
  - SHA512 and ECDSA384 examples added for F28P55x in crypto libs.
  - Alternative extended clock stretching I2C controller and target examples on F28P551x
  - Empty CLA driverlib example
  - SPI/DMA loopback driverlib example on F2837xD, F2837xS, and F2807x

- MCUBoot based SecureBoot for F28P55x in crypto libs.-
  - mcuBoot_boot_manager(Based on opensource mcbuboot bootloader)
  - mcuBoot_test_app (integrated with mcuboot signing)
- MCUBoot based Secure Firmware Update in crypto libs.-
  - mcuBoot_boot_manager (Based on opensource mcbuboot bootloader)
  - mcuBoot_test_app (integrated with mcuboot signing)
  - flash_programmer_application_UART (application to flash the new firmware. Part of firmware update process)

6. Fixed In This Release

Key	Summary	Severity
C2000WAREPKG-809	Says C2000 CGT 22.6.3 is unavailable but it is there, potential ID mismatch	S2-Major
C2000WAREPKG-801	[XBAR] Duplicate MUX configuration values in xbar.h - Lines 680 & 684	S1-Critical
C2000WAREPKG-799	F28P65x Dual-Core Bitfield Examples Not Correct	S3-Minor
C2000WAREPKG-693	C2000WARE: HRPWM Example9	S2-Major
C2000SYSCFG-484	SysCtl_setEMIF1ClockDivider() generated with raw integer instead of enum	S3-Minor
C2000SYSCFG-474	TMS320F28P550SG: F28P550 ADC reference configuration in sysconfig	S2-Major
C2000SYSCFG-463	CLB output override not supported for CLB tile 3 and 4 in P551	S1-Critical
C2000SYSCFG-331	CLB Tile Design settings reset when changing tile Name	S3-Minor
C2000DRIVERS-3364	TMS320F28P650SK: PLL initialization issue	S2-Major

7. Known Issues And Limitations

There are few limitations and known issues associated with this release

7.1. Limitations

- Many examples for the libraries available in the “dsp” and “math” folders are not ported to F28P65x and will fail to compile for F28P65x. For FPU and FPUFASTRTS libraries, the "DEVICE" parameter has to be changed to "f28P65x" in .cproject and .project to port them for F28P65x
- Many examples for the libraries available in the “dsp” and “math” folders are not ported to F28P551x. Users can refer the examples available for F28P55x and migrate to F28P551x using the SysConfig tool's Switch option
- F28P55x USB Host Mode examples - These cannot be run on the Control Card. The device gets detected but it is not enumerated successfully.
- Few examples cannot be exercised on ControlCARD due to specific peripherals/pin outs not being available on that platform - CM-UART echoback, CM-DCAN, Ethernet Rev-MII . HIC
- A few of the ECC related driverlib APIs and the message, Rx FIFO and Error Counter Status APIs have not been validated on F2838x.
- CM – Ethernet Driver: Auxiliary Time Stamp, Pulse Per Second Output features has not been validated on the driver.

7.2. New Known Issues

Key	Summary	Workaround	Severity
C2000SYSCFG-492	EPWM_disableCounterCompareShadowLoadMode gets overwritten by EPWM_setCounterCompareShadowLoadMode		S2-Major
C2000SYSCFG-479	CMPSS MUX Select with CPU2		S2-Major

7.3. Existing Known Issues

Key	Summary	Workaround	Severity
VCUDSP-40	Viterbi Decoder Example Not Working (VCU0)		S2-Major
VCUDSP-31	Build issues with VCU0 examples when using newer versions of compiler	None.	S2-Major
F2838x-159	FlashPumpSemaphoreRegs structure is missing from header file	The register is present in the IPC section. In case of bitfields, it is not possible to expose the same register in 2 different structs. It would cause linker errors. It is recommended to use the register from the IPC struct. Another workaround is to define FlashPumpSemaphoreRegs as a macro of Cpu1toCpu2IpcRegs in one of the common header files #define FlashPumpSemaphoreRegs Cpu1toCpu2IpcRegs So, any legacy code that refers to the register as FlashPumpSemaphoreRegs.PUMPREQUEST gets converted to Cpu1toCpu2IpcRegs.PUMPREQUEST.	S2-Major

IQMATH-31	Indexed library will result in build failure for some legacy device projects	None	S3-Minor
IQMATH-20	script to view wave forms of example (setupDebugEnv.js) not correct	view the Dlog variables using the graph on CCS and provide the start and end address of the variables to be plotted.	S2-Major
FPUDSP-75	rfft_adc_f32, rfft_alt_f32. rfft_alt_f32_windowed failures due to tolerance mismatch	rfft_adc_f32, rfft_alt_f32 – examples will indicate fail = 1, pass = 513. 1 failure is due to phase being computed as 0 while we are looking for 6.28 (is same as phase 0). rfft_alt_f32_windowed - pass 334/fail 51 - the tolerance have to be made 0.012 to make the fail to pass.	S3-Minor
FPUDSP-74	cfft_f32_windowed,irfft_f32, rfft_adc_f32_windowed,rfft_alt_f32_unaligned examples do not work	cfft_f32_windowed - use the RAM_ROMTABLES configuration. irfft_f32 - use the RAM_EABI configuration. rfft_alt_f32_unaligned - use the RAM_EABI configuration.	S3-Minor
FPUDSP-73	Benchmarks results in the User guide do not match with measured values		S2-Major
FPDSP-46	Magnitude of FFT is being scaled compared to the input waveform amplitude		S2-Major

FPDSP-45	FIR16 and FIR32 Flash configurations and FIR32_Alt example do not work	Please use the RAM_EABI configuration for FIR16 and FIR32 examples.	S3-Minor
FPDSP-43	Latest compiler tools will result in build failure for some legacy 2833x Fixed point DSP Projects	None.	S3-Minor
F28X7X-400	flash_programming_dcsm example does not work for f2837xS	Customers can refer to f2837xS flash_programming example for changes required.	S3-Minor
CLAMATH-47	Linker error with F2838x CLAMath examples when VCRC enabled due to Rev 0 ROM		S2-Major
CLAMATH-46	CLA Library and FFT functions does not support EABI. Update documentation for FFT functions.		S2-Major
C28XECAT-126	Potential WORD_ALIGN Bug for CM Examples		S3-Minor
C2000WAREPKG-781	TMS320F28P650DK: EtherCAT Slave Initialization		S3-Minor
C2000WAREPKG-679	P55x ROM.out file does not match source file		S2-Major
C2000WAREPKG-620	CLB boundary input simulator using incorrect period.		S3-Minor
C2000WAREPKG-612	BootROM Code does not align while loading symbols		S2-Major
C2000WAREPKG-573	Secure ROM .lib file does not work as expected in COFF format		S3-Minor
C2000WAREPKG-528	#pragma diag_error in F28x_Project.h switches off the errors 70,770 and 232.		S2-Major
C2000WAREPKG-480	Clocking Issue with CM EtherCAT Example		S3-Minor

C2000WAREPKG -237	indexed library leades to build issues for 28035 (2803x), 28069 (2806x), 28235 (2823x), 28335(2833x) projects	C example 2803x – delete the index library IQmath.lib and rename the coff library IQmath_coff.lib as the IQmath.lib C example 2806x – delete the index library IQmath_fpu32.lib and rename the coff library IQmath_fpu32_coff.lib as the IQmath_fpu32.lib C example 2823x – delete the index library IQmath.lib and rename the coff library IQmath_coff.lib as the IQmath.lib C example 2833x – delete the index library IQmath_fpu32.lib and rename the coff library IQmath_fpu32_coff.lib as the IQmath_fpu32.lib C++ example 2803x - delete the index library IQmath.lib and rename the coff library IQmath_coff.lib as the IQmath.lib C++ example 2806x – build fails	S3-Minor

C2000WAREPKG -198	TIREX: Build failure for 28035_IQsampleCpp	C example 2802x – delete the index library IQmath.lib and rename the coff library IQmath_coff.lib as the IQmath.lib C example 2803x – delete the index library IQmath.lib and rename the coff library IQmath_coff.lib as the Iqmath.lib C example 2806x – delete the index library IQmath.lib and rename the coff library IQmath_coff.lib as the IQmath.lib Cpp example 2833x – delete the index library IQmath_fpu32.lib and rename the coff library IQmath_fpu32_coff as the IQmath_fpu32.lib	S3-Minor
C2000SYSCFG-472	MCAN_EX3: Incorrect FIFO Size Configuration Causes Stall - Requires Update		S2-Major
C2000SYSCFG-392	Copy settings feature in epwm syscfg does not copy over phase shift value		S2-Major
C2000SYSCFG-358	TMS320F28379S: sysconfig-1.21.2_3837 & C2000Ware_5_04_00_00: Error: myEMIF10(/driverlib/emif1.js) useInterfacePins: Missing required pin signal: EM1DQM1& EM1DQM2		S2-Major
C2000SYSCFG-355	Dx RAM allocation to CPU2 is mapped to wrong address in linker cmd tool generated file		S2-Major

C2000SYSCFG-327	CMD unable to have same section names for different profiles		S2-Major
C2000SYSCFG-239	Update the CLA compiler section when we choose RAMLSx memory as CLA Program Memory		S2-Major
C2000DRIVERS-3066	F28P55x_SecureZoneCodeSymbols.lib has incorrect VCU compiler setting		S2-Major
C2000DRIVERS-2538	[DCAN] ISR entered twice for every event		S2-Major
C2000DRIVERS-2040	Reference interrupt handlers provided in driverlib can be in infinite loop	None. The interrupt handler to be updated to avoid this condition	S2-Major
C2000BROM-740	F28002x ROM CAN bootloader alternate modes point to default mode		S2-Major

8. Dependencies

This release has dependency on the following tools

- CCS v20.4.1 - This includes all the below 3 dependencies and need not be installed separately
- TI C2000 Compiler v22.6.3.LTS
- TI ARM Compiler v20.2.7.LTS
- SysConfig 1.27 or later

9. References

1. Getting Started with C2000™ Software, get an overview of Software development and various Software releases available - C2000™ Software Guide
2. C28x Software development and optimization guide – C2000™ C28x optimization Guide
3. Adding SYSCONFIG support to C2000 Driverlib Projects - E2E FAQ
4. Control Law Accelerator Software development – C2000™ CLA Software Development Guide
5. C28x Compiler v22.6.3 LTS User's Guide
6. C28x Assembler Tools v22.6.3 LTS User's Guide
7. ARM Compiler v20.2.7 LTS User's Guide
8. ARM Assembler Tools v20.2.7 LTS User's Guide

9. controlSUITE To C2000Ware Transition Guide
10. Programming TMS320x28xx and 28xxx Peripherals in C/C++

10. Technical Support

C2000 microcontroller e2e forum